

AWS Exercise

Q1. List out the types of instance base on the pricing model and write a brief about your understanding about it.

On-Demand instances:

On-Demand instances let you pay only for what you use without any long-term commitment. They are used for workloads that don't have a fixed usage pattern. I feel this option is best when flexibility matters more than cost savings though it can be expensive if used continuously.

Reserved Instances & Savings Plans:

These options are meant for workloads that run regularly over a long period. By committing for one or three years you get significant cost savings compared to On-Demand pricing. In my understanding they are ideal for stable applications like databases or backend services.

Spot Instances:

Spot Instances are much cheaper because they use unused AWS capacity but they can be interrupted at any time because of this they are best for tasks like background jobs. I see Spot Instances as a great cost-saving option if the application can handle interruptions.

Dedicated Instances:

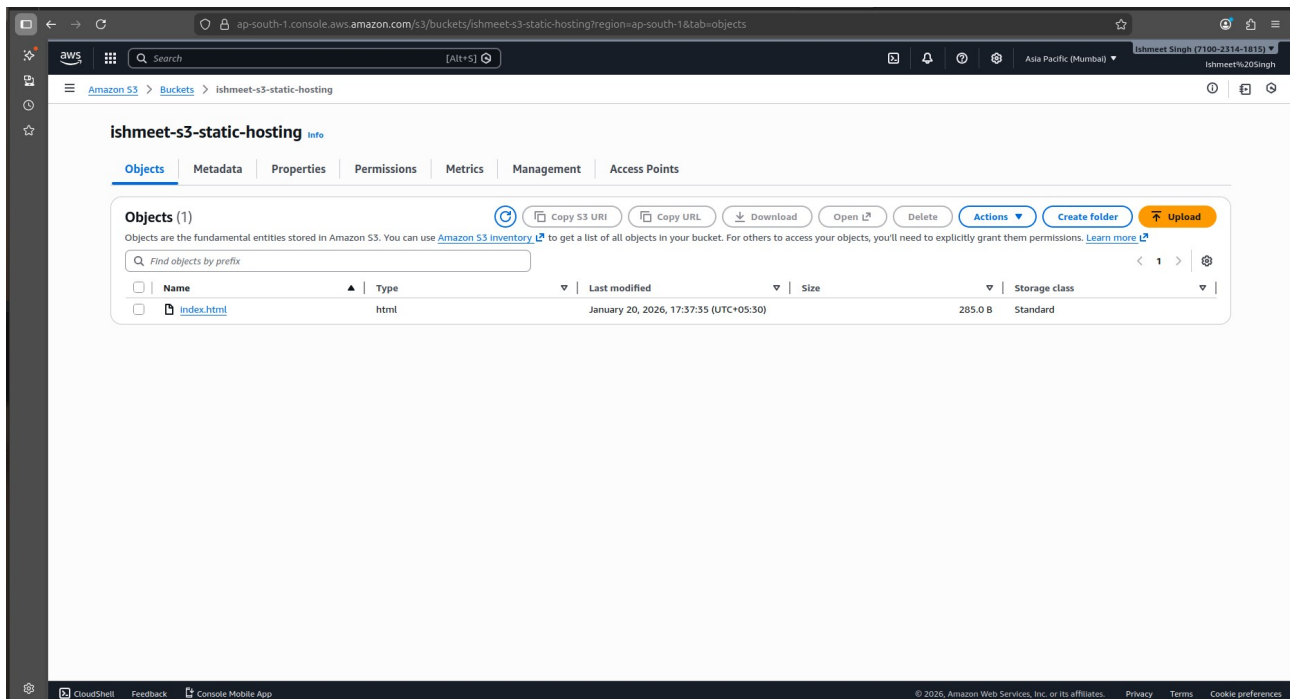
Dedicated Instances run on hardware meant for a single customer offering better isolation than shared instances and other customer can rent machine on that hardware. They are mainly used to meet compliance or security needs rather than to reduce costs.

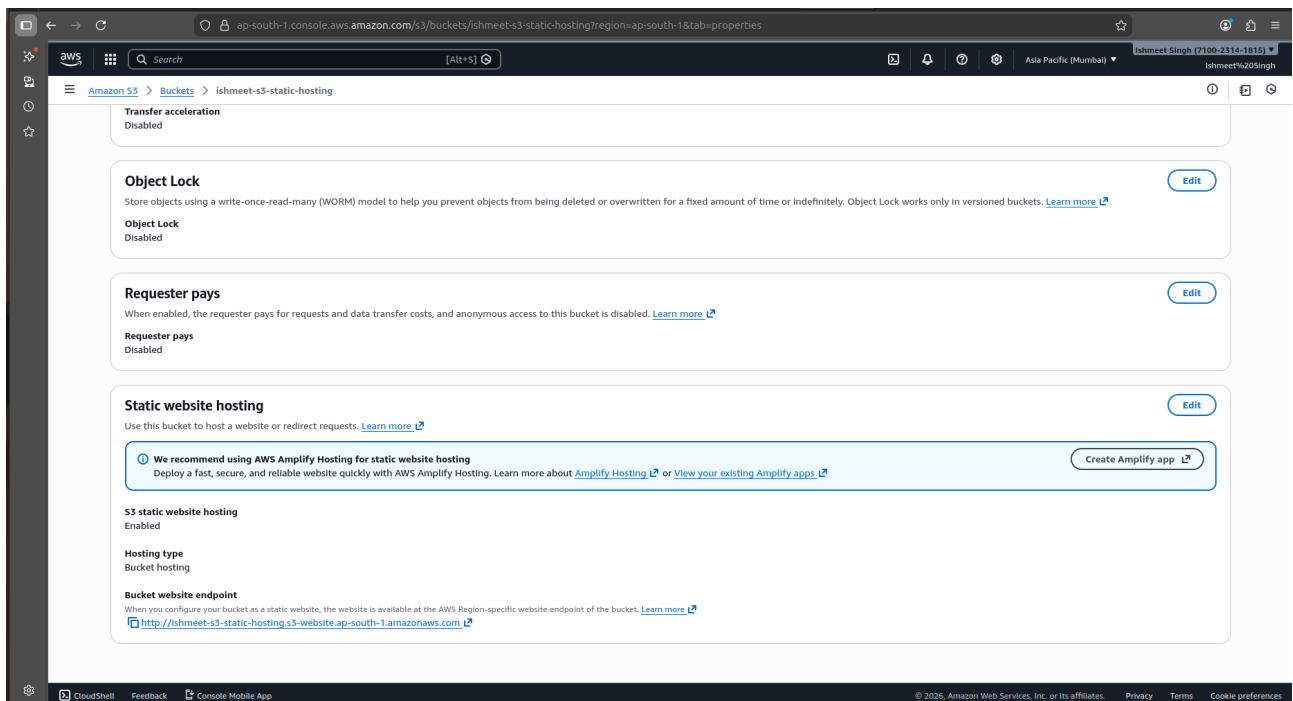
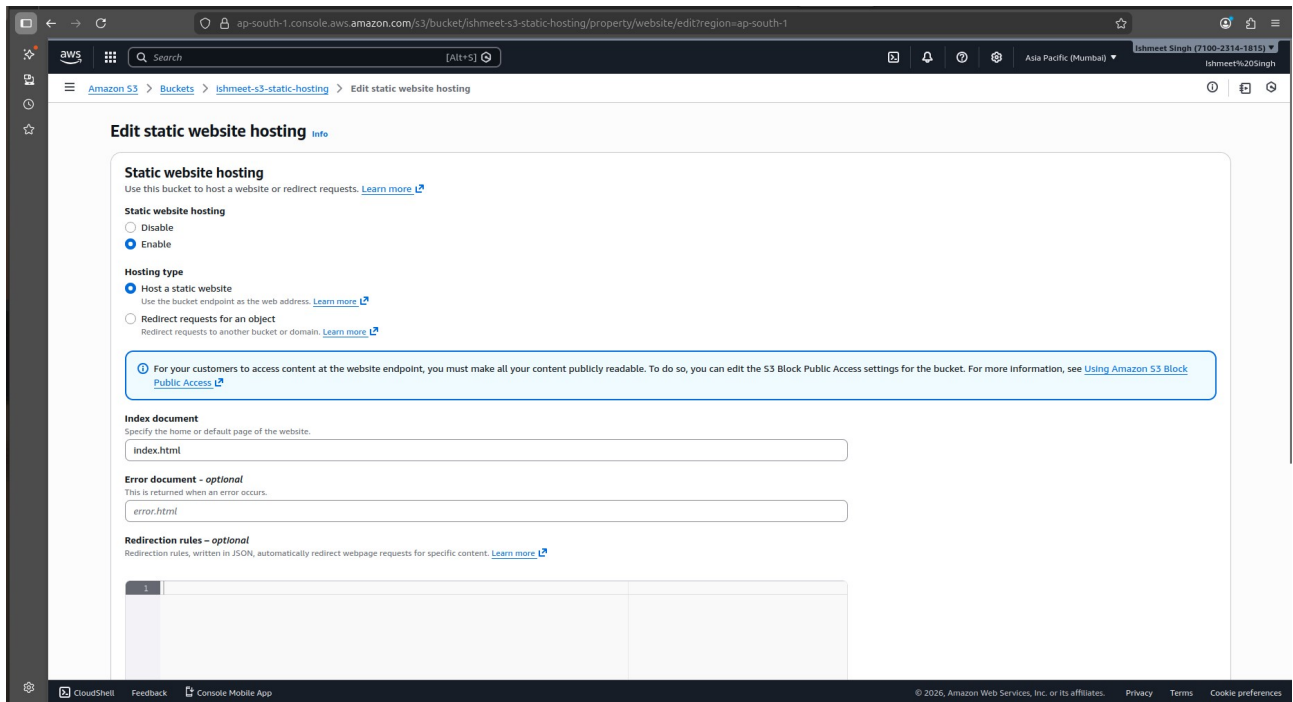
Dedicated Hosts:

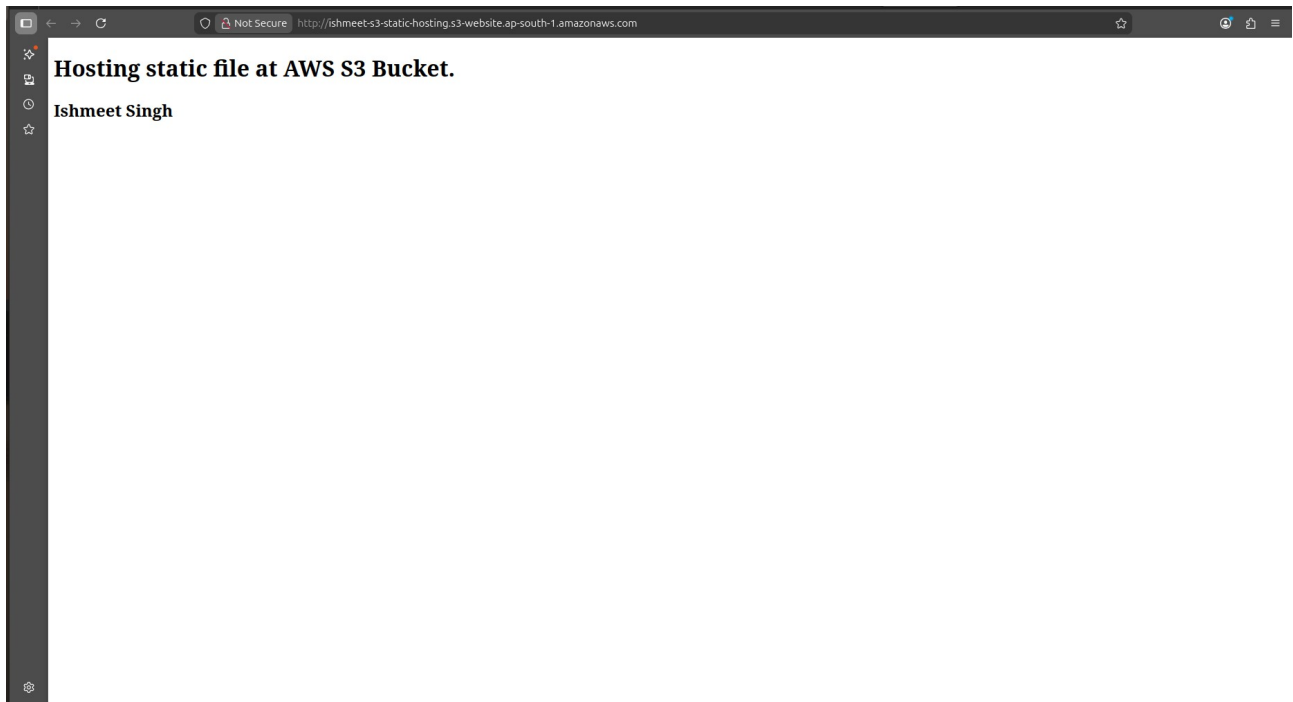
Dedicated Hosts allow full control over a physical server and are mainly used for hardware based software licensing and strict compliance needs. They are costly but provide the highest level of control. This is especially useful when you have software licenses that are tied to physical hardware such as certain enterprise databases or operating systems.

Q2. Host a static website in S3

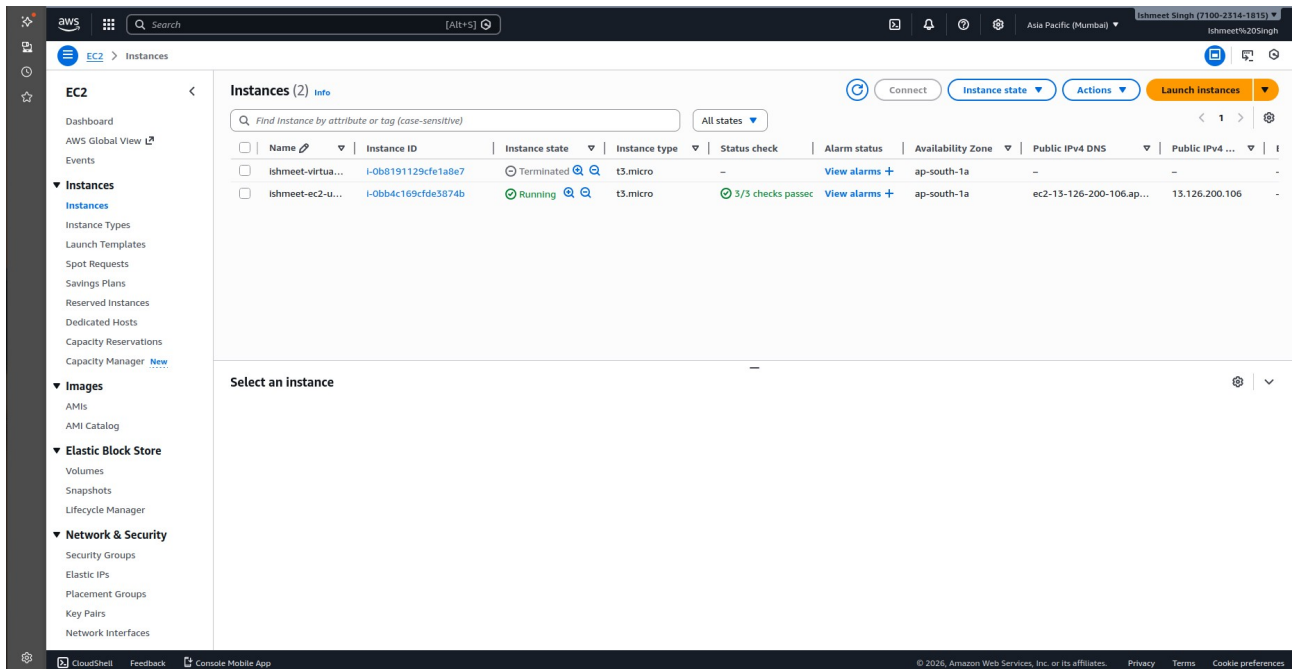
```
ishmeet-singh@TTNPL-ishmeet:~/aws-exercise$ cat index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document Title</title>
</head>
<body>
  <h1>Hosting static file at AWS S3 Bucket.</h1>
  <h2>Ishmeet Singh</h2>
</body>
</html>
ishmeet-singh@TTNPL-ishmeet:~/aws-exercise$
```







Q3. Launch an Ubuntu EC2 instance on AWS, with 10GB root volume, and SSH from your local machine using the private key.



EC2

Dashboard

AWS Global View

Events

Instances

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Launch Templates

Spot Requests

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Placement Groups

Key Pairs

Network Interfaces

Instance summary for i-0bb4c169cfde3874b (ishmeet-ec2-ubuntu) Info

Updated less than a minute ago

Instance ID

i-0bb4c169cfde3874b

IPv6 address

—

Hostname type

IP name: ip-172-31-35-201.ap-south-1.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

13.126.200.106 [Public IP]

IAM Role

—

IMDSv2

Required

Operator

—

Public IPv4 address

13.126.200.106 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-35-201.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-028f7dd47bff51daa | [open](#)

Subnet ID

subnet-07ea6d5b6fb40b793 | [open](#)

Instance ARN

arn:aws:ec2:ap-south-1:710023141815:instance/i-0bb4c169cfde3874b

Private IPv4 addresses

172.31.35.201

Public DNS

ec2-13-126-200-106.ap-south-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses

—

AWS Compute Optimizer finding

[Opt-in to AWS Compute Optimizer for recommendations.](#) | [Learn more](#)

Auto Scaling Group name

Managed

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance details Info

AMI ID

ami-02b8269d5e85954ef

Monitoring

disabled

Platform details

Linux/UNIX

CloudShell

Feedback

Console Mobile App

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Storage

Tags

▼ Root device details

Root device name

/dev/sda1

Root device type

EBS

EBS optimization

enabled

▼ Block devices

Filter block devices

☒

Volume ID

☒

Device name

☒

Volume size (GiB)

☒

Volume State

☒

Attachment status

☒

Attachment time

☒

Encrypted

☒

KMS key ID

vol-02c65cbe11087dd2b

/dev/sda1

10

In-use

Attached

2026/01/20 22:41 GMT+5:30

No

—

Volume monitoring (1)

☐ Alarm recommendations ?

[Investigate with AI - new](#)

1h

3h

12h

1d

3d

1w

Custom

UTC timezone

[Explore related](#)

Stalled I/O Check

None

Average read latency (ms/op)

1.24

Average write latency (ms/op)

1.95

Read throughput (KiB/s)

792.987

Write throughput (KiB/s)

815.282

Read operations (Ops/s)

21.7

Write operations (Ops/s)

8

Average queue length (Operations)

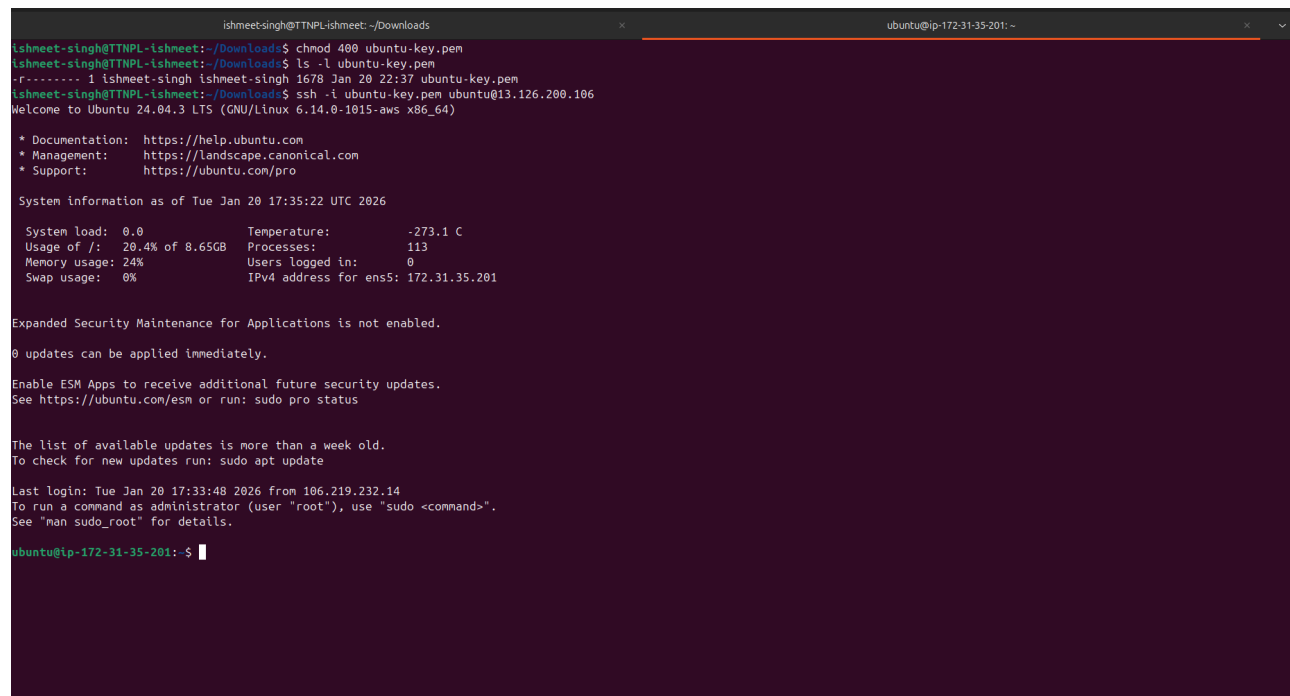
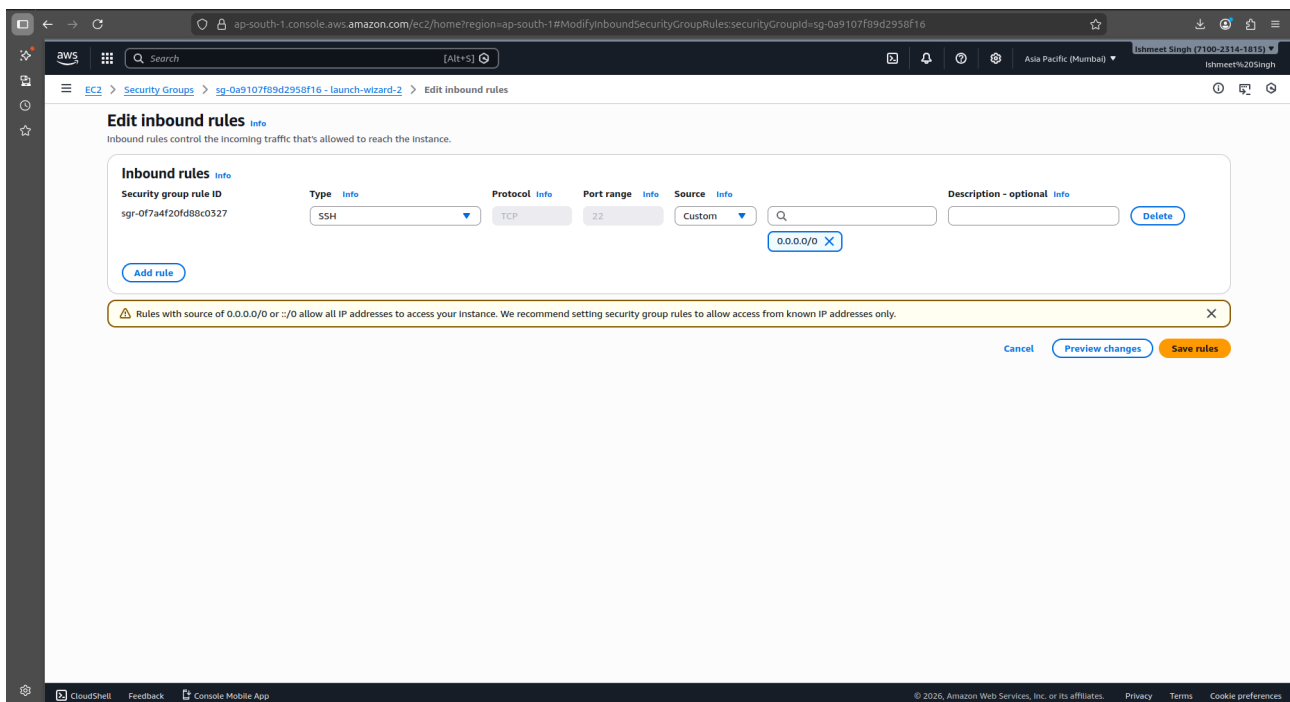
Count 0.08

CloudShell

Feedback

Console Mobile App

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Q4. Install nginx package in the above server and access this page from your local browser using a domain name instead of IP address of the server.

```
ubuntu@ip-172-31-35-201:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 86 not upgraded.
Need to get 564 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43.4 kB]
Get:2 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 564 kB in 0s (24.3 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 71735 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.5_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.5) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.5_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.5) ...
Setting up nginx-common (1.24.0-2ubuntu7.5) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.5) ...
* Upgrading binary nginx
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

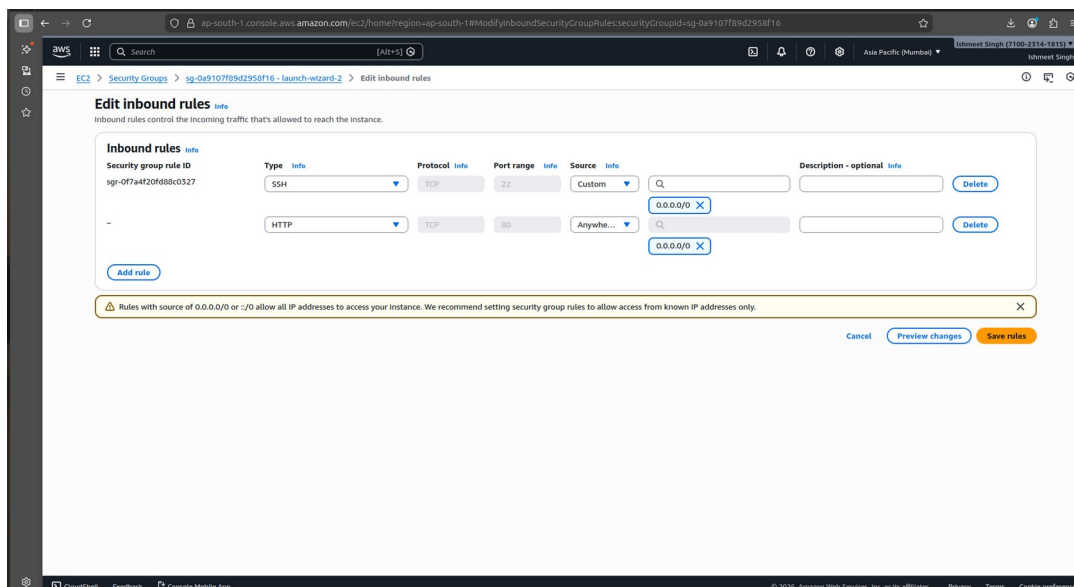
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-35-201:~$
```

```
ubuntu@ip-172-31-35-201:~$ sudo systemctl start nginx
ubuntu@ip-172-31-35-201:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-172-31-35-201:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-01-20 17:55:47 UTC; 2min 33s ago
     Docs: man:nginx(8)
    Main PID: 2083 (nginx)
      Tasks: 3 (limit: 1008)
     Memory: 2.4M (peak: 5.1M)
        CPU: 28ms
    CGroup: /system.slice/nginx.service
            └─2083 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─2085 "nginx: worker process"
                  └─2086 "nginx: worker process"

Jan 20 17:55:47 ip-172-31-35-201 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 20 17:55:47 ip-172-31-35-201 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
```



```
GNU nano 7.2 /etc/hosts
127.0.0.1 localhost
127.0.1.1 TTNPL-ishmeet

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe80::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
13.126.200.106 mynginx-ishmeet.local
```

[Read 10 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-T To Bracket M-Q Previous ^B Back
^X Exit ^R Read File ^N Replace ^U Paste ^J Justify ^/_ Go To Line M-E Redo M-G Copy ^Q Where Was M-W Next ^F Forward

